

Approaches for Orthology, Annotation, and Cross-Species Comparisons

Resources for Plant Sciences: Data Integration and Interpretation Tools

Tutorial Series · Systems biology, Multi-omics Integration and Modelling

GitHub: [NIB-SI/ECCB2026_T12](#) · 2026

Agenda

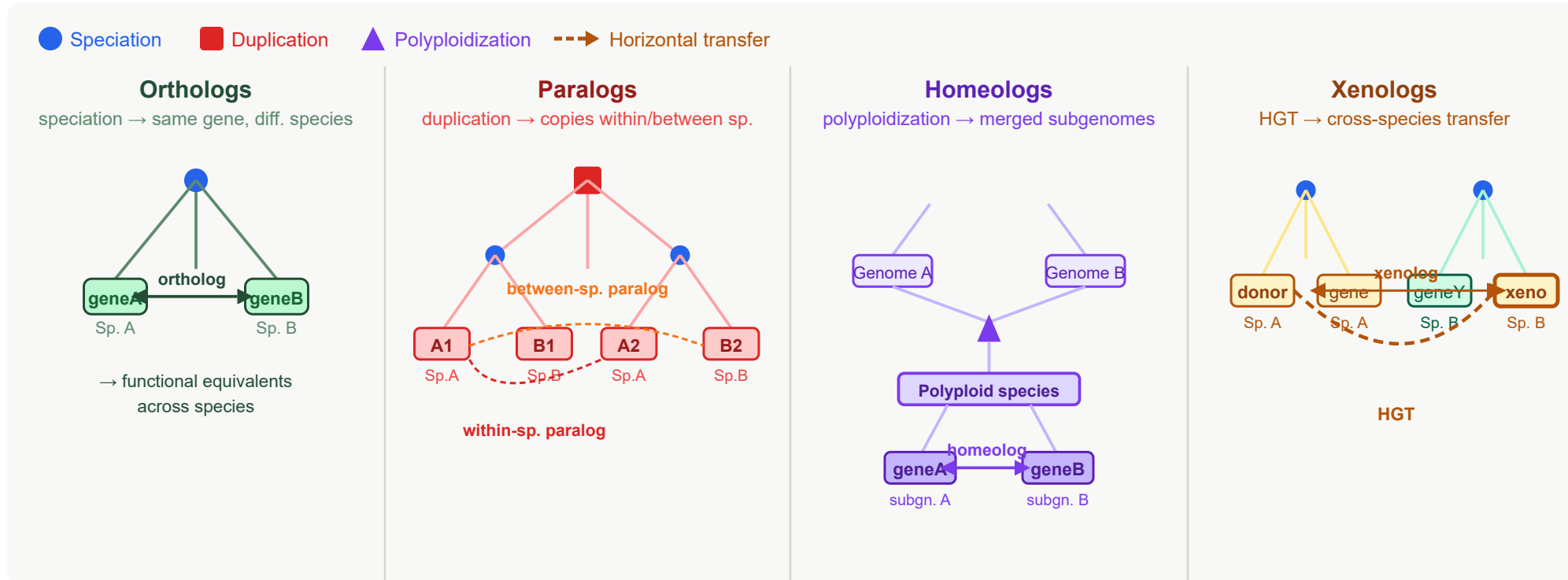
1. **Introduction**
2. **Plant genomes**
3. **Why orthology?**
4. **Data sources**
5. **Tools**
6. **Integration & annotation**
7. **Workflow & summary**

Introduction

Orthology and sequence similarity

What is Orthology?

Homology = evolutionary relatedness between sequences.



Key principle: Orthologs tend to retain function across species — this is the basis for knowledge transfer from well-studied model organisms to less-studied crops

Why Do We Need Multiple Methods?

No single tool is best in all situations. Each captures a different signal:

Type	Objective	Species/genome version addition?	Stable?	E.g. approach
Pairwise sequence similarity	Mutual best hits between two species	Each pair runs independently	Partially — ties broken arbitrarily	RBH BLAST+
Graph clustering (all-vs-all)	Cluster all genes into orthogroups	Requires full rerun	No — whole graph and clusters rebuilt	OrthoFinder
Pre-computed database	Query against fixed reference orthogroups	Database is frozen at release	N/A	OrthoDB

Why Do We Need Multiple Methods?

Type	Objective	Species/genome version addition?	Stable?	E.g. approach
Gene tree reconciliation	Reconciling gene tree with species tree	Requires full rerun	Tree topology changes with new taxa	Compara, PLAZA, FastOMA
Synteny	Identify collinear gene blocks	Yes	Pairwise: Yes. Multi-way: No — changes block detection	JCVI/MCScanX
Functional classification	Assign functional bins to protein sequences	Yes	Depends on the reference dataset version	MapMan4/Mercator

Strategy: run all methods independently, then **integrate and filter** by priority and consensus — giving the most reliable set of ortholog pairs

Plant genomes

Complexity & Polyploidy

Plant Genomes

Plants have some of the most **complex and dynamic genomes** of any organism:

- **Massive size range:** 135 Mbp (*Arabidopsis*) → 16 Gbp (wheat)
- **High repeat content:** up to 80–90% transposable elements in large genomes
- **Frequent polyploidy:** whole-genome duplications (WGD) throughout plant evolution
- **Ongoing gene family expansion:** NBS-LRR resistance genes, cytochrome P450s, ERFs and other TFs
- **Structural variation:** extensive inversions and translocations between closely related species

Plant Genomes

Consequences for bioinformatics:

- Many-to-many ortholog relationships
- High risk of confusing paralogs with orthologs
- Synteny breaks down across large evolutionary distances

Why it matters:

- Most of study species are polyploids
- *Arabidopsis* is misleadingly "simple"

Polyploidy

Autopolyploidy — duplication of the same genome

$A A \rightarrow A A A A$ (same species, chromosome doubling)

Allopolyploidy — hybridisation of two species + chromosome doubling

$A A + B B \rightarrow A A B B$ (hybrid, then doubling)

Ancient / Paleopolyploidy — old WGD followed by diploidisation

- ~70% of flowering plant species have a polyploid history
- Core eudicots share an ancient γ WGD
- Arabidopsis has undergone at least 3 WGDs
- Efficient diploidisation restored a diploid-like genome

Consequences for Orthology

Gene fates after WGD:

- **Subfunctionalisation** — each copy retains part of the ancestral function
- **Neofunctionalisation** — one copy evolves a new function
- **Gene loss (fractionation)** — one copy silenced and eventually deleted
- **Dosage balance** — transcription factors and kinases often retained in pairs

Ploidy Diversity Across Plant Genomes

Species	Ploidy	2n	Key point
<i>Arabidopsis thaliana</i>	Diploid	10	Ancient WGDs; model of diploidisation
<i>Vitis vinifera</i>	Diploid	38	Slowly evolving genome; preserves ancestral genome structure
<i>Malus domestica</i>	Diploid	34	Lineage-specific WGD; extensive retained duplicates
<i>Solanum lycopersicum</i>	Diploid	24	Solanaceae WGD; representative crop genome
<i>Solanum tuberosum</i>	Segmental allotetraploid	48	Substantial heterozygosity and historical introgression
<i>Triticum aestivum</i>	Allohexaploid	42	Three subgenomes (AABBDD); major cereal crop
<i>Fragaria</i> × <i>ananassa</i>	Octoploid	56	Four subgenomes (4 diploid progenitors); complex allopolyploid origin

Why bother?

Context and motivation

Exploiting the untapped potential of fruit tree wild diversity for sustainable agriculture

FruitDiv · Horizon Europe · fruitdiv.eu

The problem: Current fruit production relies on a handful of **elite cultivars** — high yield and shelf life, but declining resistance to pests and disease and vulnerable to climate change

Crop Wild Relatives (CWR) of pome (*Malus*, *Pyrus*) and stone (*Prunus*) fruits hold untapped genetic diversity:

- Pest and disease resistance
- Drought and salinity tolerance
- Adaptability to marginal lands and fluctuating climates

Cross-species orthology — to unlock stress tolerance traits from wild relatives and transfer functional knowledge into breeding programmes, we must first know which genes are equivalent across species

Accelerated Development of multiple-stress tolerant Potato

ADAPT · Horizon Europe · adapt.univie.ac.at

The problem: Potato is a non-model polyploid staple crop and a extremely sensitive to environmental stress:

- Heat + drought → yield reduction, quality losses
- Flooding → entire harvest lost within 1–2 days

Cross-species orthology links **conserved signalling pathways** from model species to potato

The Knowledge Gap: Model to Crop

The challenge: Most molecular knowledge comes from model organisms — not from crops

Arabidopsis thaliana

Fully annotated
40+ years of research
Thousands of mutants
Rich ontology annotations
Deep stress response data
High-quality genome

Cultivars

Partially annotated
Limited functional data
Few or no mutants
Many genes = "unknown"
Stress mechanisms not fully understood
T2T genomes are more an exception than a rule

The Knowledge Gap: Model to Crop

Orthology-based annotation transfer bridges this gap:

1. Identify ortholog of Arabidopsis stress gene in potato / apple / peach
2. Transfer ontology terms, pathway membership, knowledge graph node assignment
3. Prioritise candidates for experimental validation
4. Feed into multi-omics research and pre-breeding programmes

Without reliable orthologs, functional genomics in crops starts from zero every time

Data Sources

The genomes, databases, and reference resources

Databases & Data Sources

Type	Database	Focus
Genome sources	GDR	Rosaceae genomes and versions
	SpudDB / UniTato	Potato genomes and versions
	Sol Genomics	Solanaceae genomes and versions
	Grapedia	Grapevine genomes and versions
	Phytozome	Multi-species plant genomes
Orthology DBs	Ensembl Plants	Pre-computed homologies
	PLAZA	Plant orthogroups
	OrthoDB	Hierarchical orthogroups (HOGs)

Note: Genome versions and ID formats differ across databases for the same species
Primary/representative transcripts (longest isoform) are commonly used throughout bioinfo pipelines

Orthology Tools

Complementary methods

OrthoFinder

Emms & Kelly (2019) *Genome Biology*

Two modes: within-species (paralogs, *genome versions* and *cultivars*); between-taxa (orthologs)

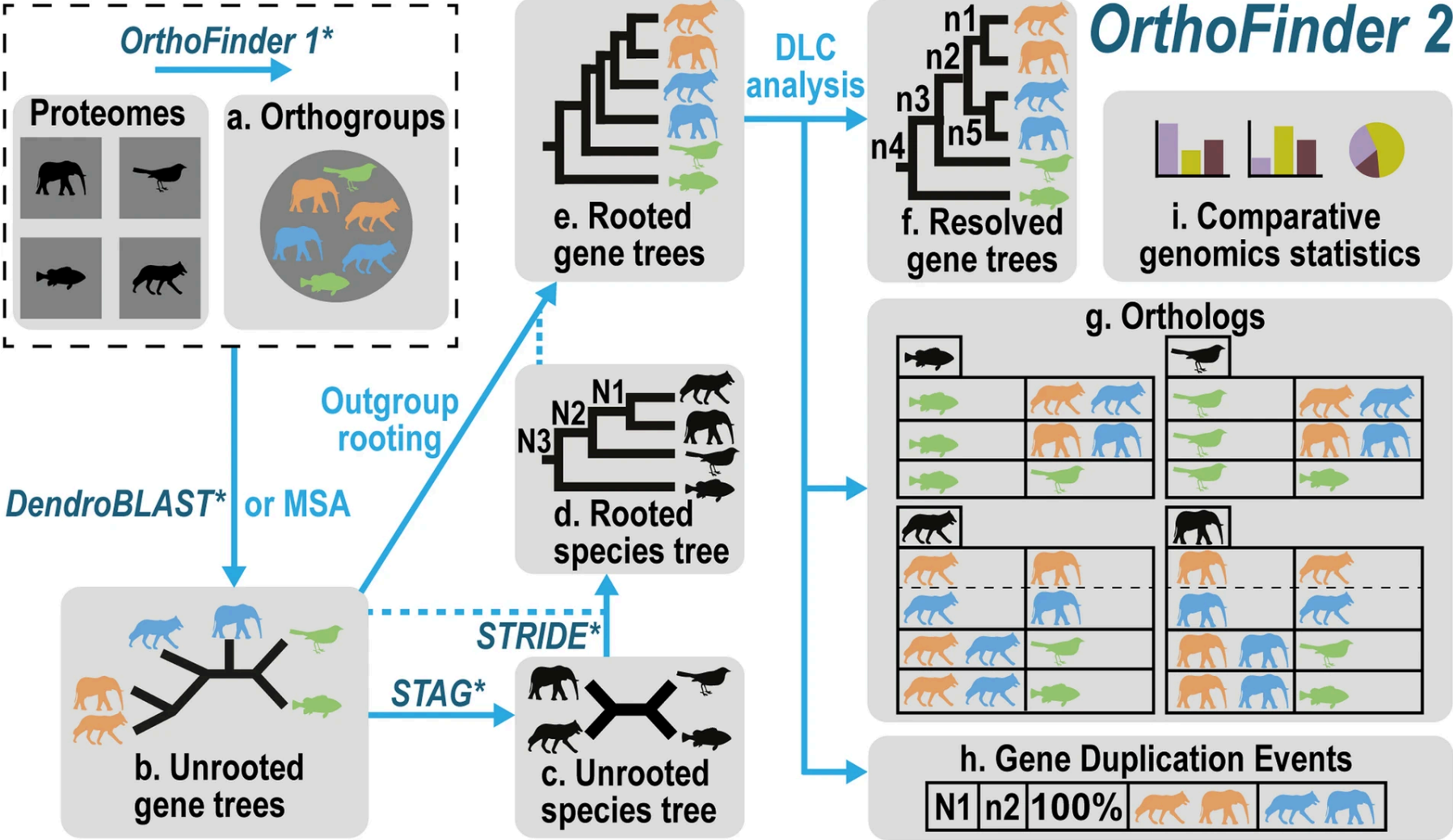
Standard workflow

- DIAMOND or MMseqs2 (recommended, although BLAST+ can be used instead)
- MCL graph clustering algorithm
- FastME

MSA workflow

- Multiple sequence alignment program: MAFFT (recommended), Muscle
- Tree inference program: FastTree (recommended), IQTREE (takes a very large amount of time to run on a reasonable sized dataset), raxml

OrthoFinder



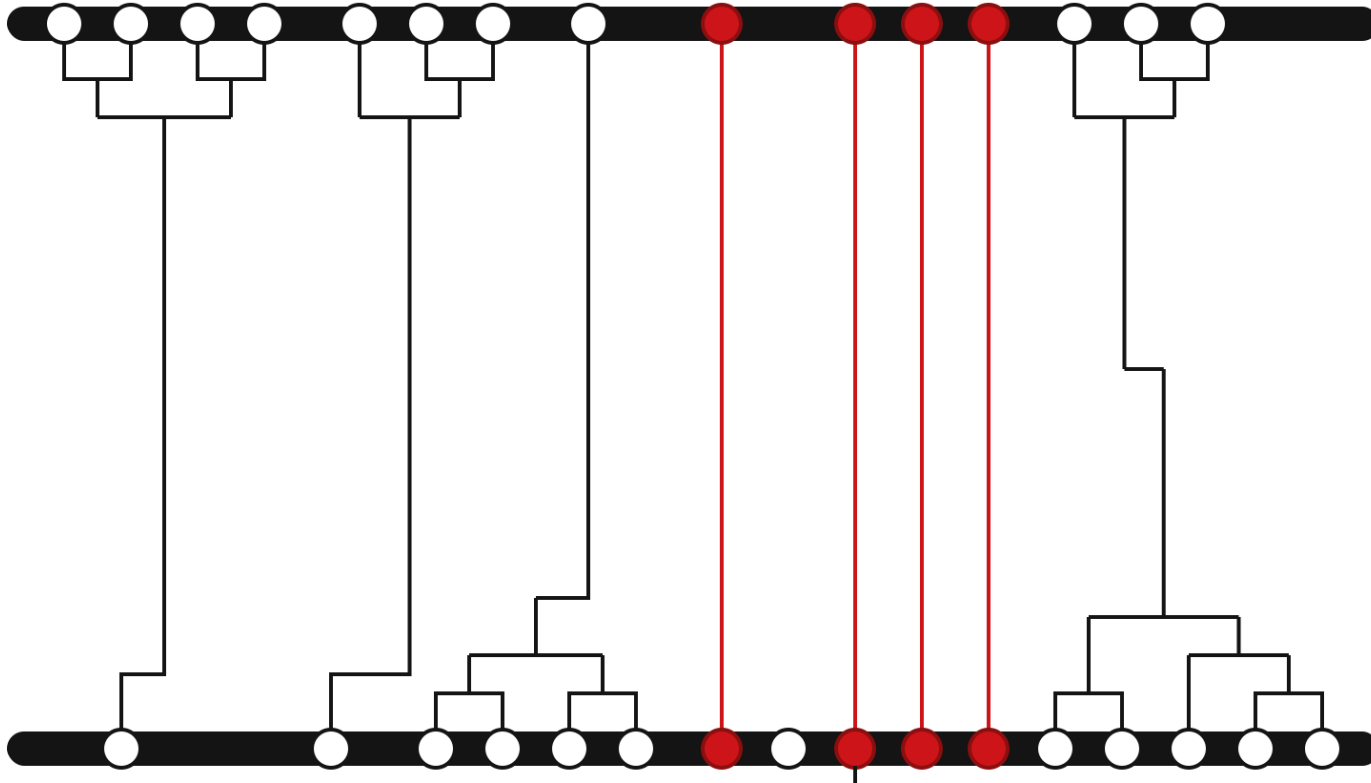
OrthoFinder

Approach: DIAMOND ultra-sensitive + MCL clustering + IQ-TREE gene trees

```
orthofinder -f "$input_dir" \  
            -M msa \  
            -S diamond_ultra_sens \  
            -T iqtree \  
            -t "$THREADS" \  
            -n orthofinder \  
            -o "$output_dir" 2>&1 | tee -a "$LOGFILE"
```

Sanity check: parameter tuning based on different *ath* genome versions

RBH (Reciprocal Best Hits)



Goodstadt, L. (2009) Slideserve

Note: Reciprocal best hits are not a logically sufficient condition for orthology (arXiv:0706.0117)

RBH (Reciprocal Best Hits)

Approach: Bidirectional BLAST → keep only pairs that are each other's top hit: reciprocals

1. Download BLAST from <https://ftp.ncbi.nlm.nih.gov/blast/> and install locally (R) and add path to blast /bin or add Bio.Blast.Applications module (Py)
2. e.g. in R: install dependencies

```
BiocManager::install(c("Biostrings", "GenomicFeatures", "GenomicRanges",  
"Rsamtools", "IRanges", "rtracklayer", "biomaRt"))
```

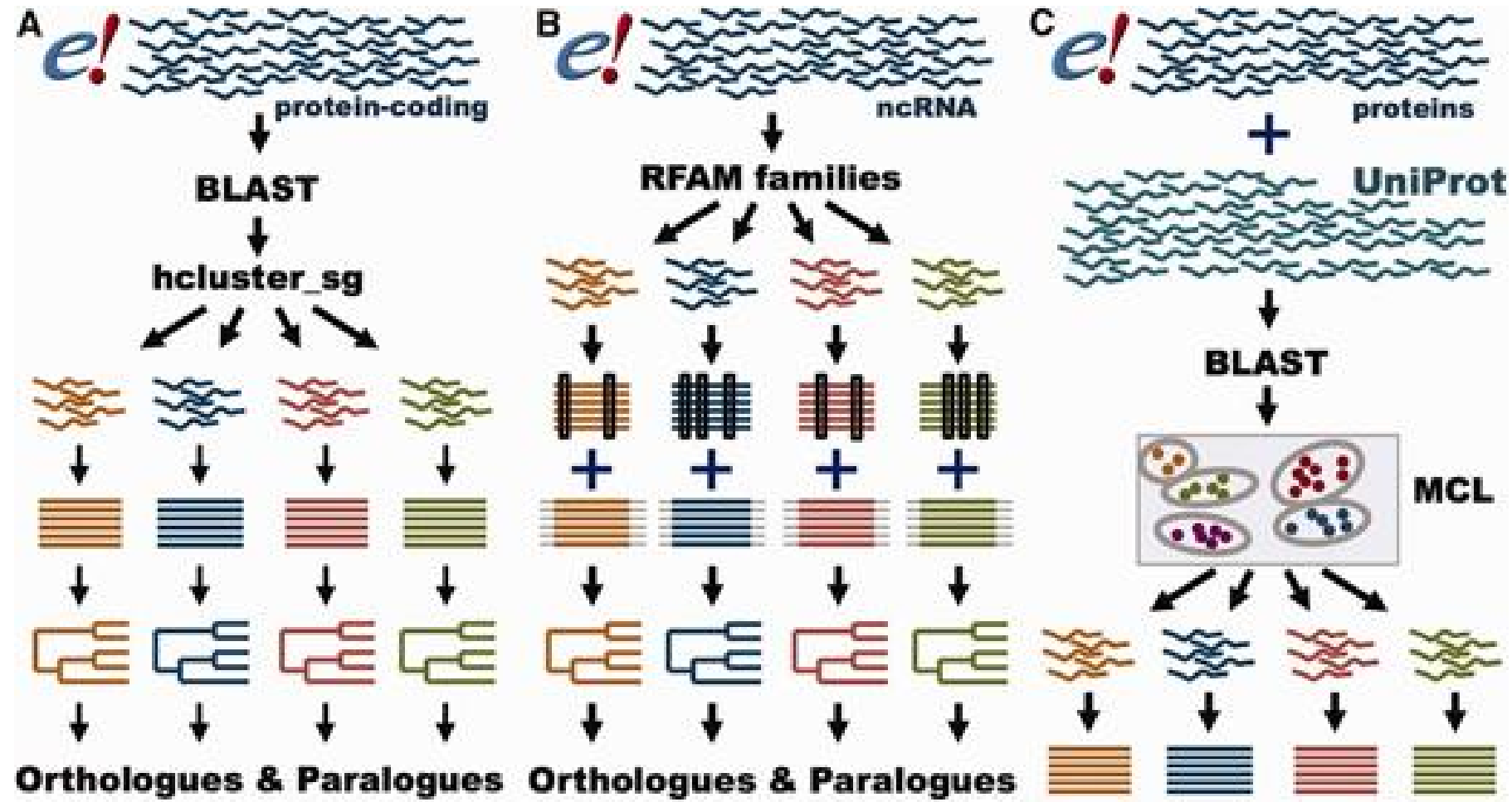
3. e.g. in R: install `metablastr` and `orthologr` packages

```
devtools::install_github("HajkD/metablastr")  
devtools::install_github("HajkD/orthologr")
```

4. e.g. in R: run in R Console to see if it is working

```
system("blastp -help")
```

Ensembl Compara



Ensembl Compara

Herrero et al. (2016) *Genome Biology* · Dyer et al. (2025) *Nucleic Acids Research*

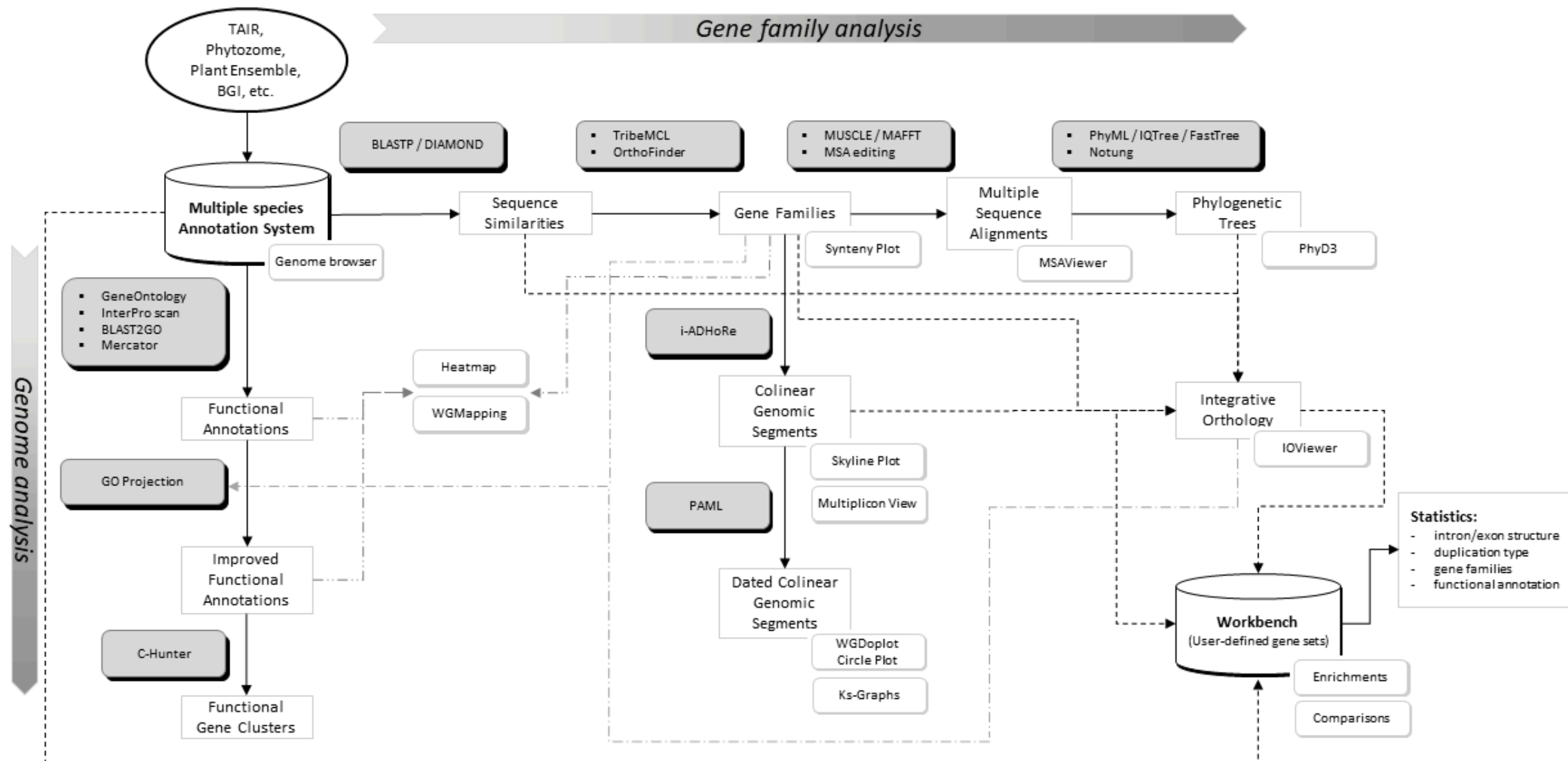
<ftp.ebi.ac.uk/ensemblgenomes/pub/plants/release-X/tsv/ensembl-compara/homologies/>

Download both $X \rightarrow Y$ and $Y \rightarrow X$ tables to obtain all gene pairs

Approach: HMM (TreeFam) family classification → BLAST + hcluster_sg (orphans) → MSA (M-Coffee until 2025 / MAFFT for large) → TreeBeST [5 trees: ML protein, ML codon, NJ p/dN/dS → consensus] → reconciliation vs Ensembl species tree/NCBI taxonomy-based species tree [mapping the gene tree onto the species tree]

High-confidence filters:

- $GOC \geq 50$ [at least 2/4 flanking genes conserved]
- $WGA \geq 50$ [LastZ/BlastZ genomic region alignment]
- $\% \text{ identity} \geq 25$ [identical AA in pairwise alignment]



PLAZA

Van Bel et al. (2022) *Nucleic Acids Research*

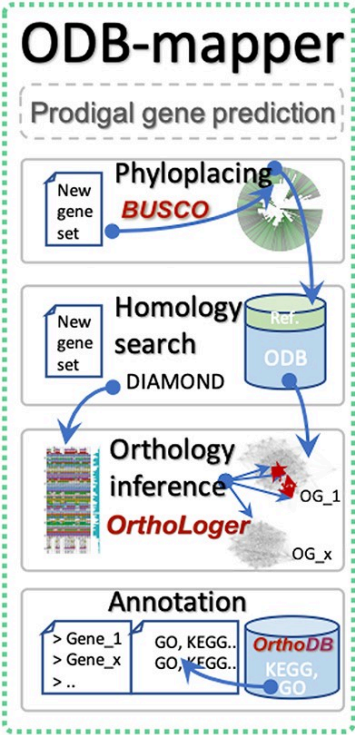
Approach: DIAMOND + Tribe-MCL (families) → OrthoFinder (sub-families) → MAFFT → FastTree
[Notung 2.6 — roots trees, infers speciation/duplication nodes via Duplication/Loss Score]

Speciation nodes → orthologs · Duplication nodes → paralogous

Code	Type	Basis
TROG	Tree-based ortholog	Position in reconciled gene tree — most rigorous
ORTHO	Sequence cluster	MCL clustering — includes orthologs <i>and</i> paralogous
BHIF	Best hit + inparalogous	RBH-like — groups gene with best hit + recent duplicates
ANCHOR	Syntenic anchor	Collinear block from i-ADHoRe — supports WGD detection

Genome evolution: **i-ADHoRe** — collinear blocks = multiplicons → PAML + CLUSTALW — aligns collinear CDS, calculates Ks (synonymous substitution rate) → peaks in Ks distribution date WGD events

OrthoDB



OrthoLogger



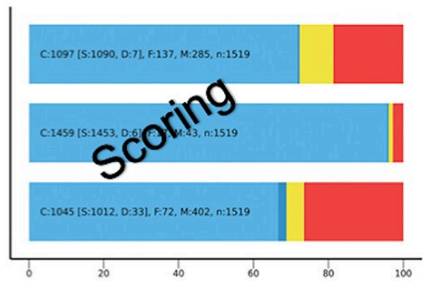
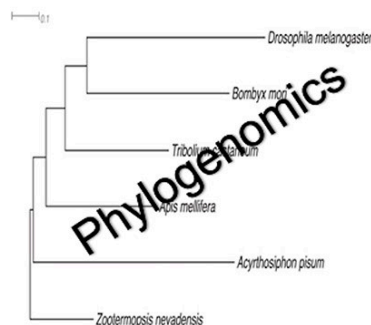
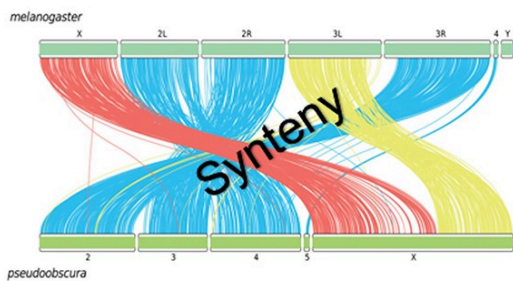
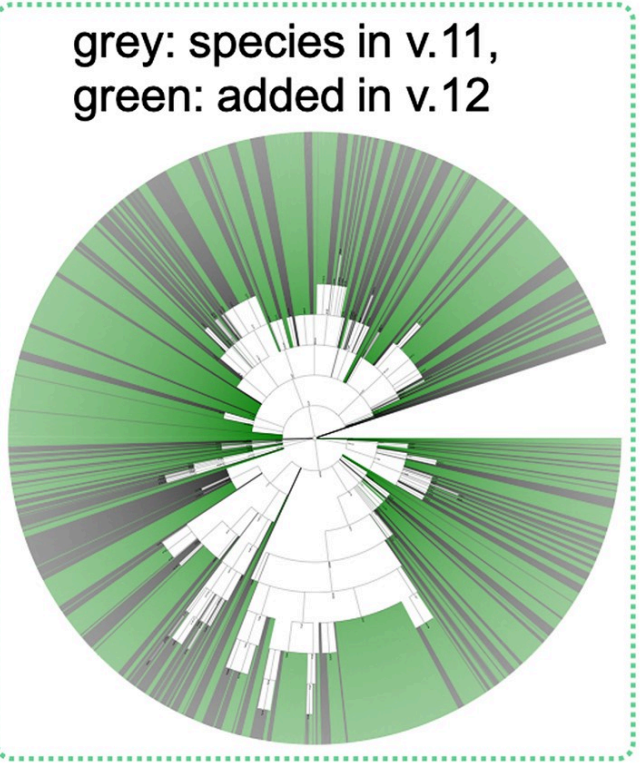
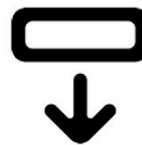
Docker & Apptainer
containers
on Bioconda &
BioContainers.pro

OrthoDB



OrthoDB API
for Python & R

BUSCO



	v10	v12 update
Eukaryota	67	109
Bacteria	83	332
Archaea	16	29

OrthoDB / OrthoLogger

Kuznetsov et al. (2023) OrthoDB v11 · Tegenfeldt et al. (2025) OrthoDB v12 *Nucleic Acids Research*

Design philosophy

- Balances *sensitivity* and *specificity*
- OrthoFinder / OrthoMCL favour inclusivity → higher sensitivity but broader families (orthologs + paralogs mixed)
- OrthoDB aims for accuracy of functional inference

OrthoLogger pipeline

- Homology: `MMseqs2` or `DIAMOND`
- Low-complexity masking: `segmasker` (NCBI Blast)
- Redundancy reduction: `cd-hit`
- Ortholog pairs: BRH between each species pair (proxy for gene tree reconciliation)
- Clustering: `BRHCLUS`

- **v12**: hierarchical mode guided by species tree → improves scalability and consistency across taxonomic levels

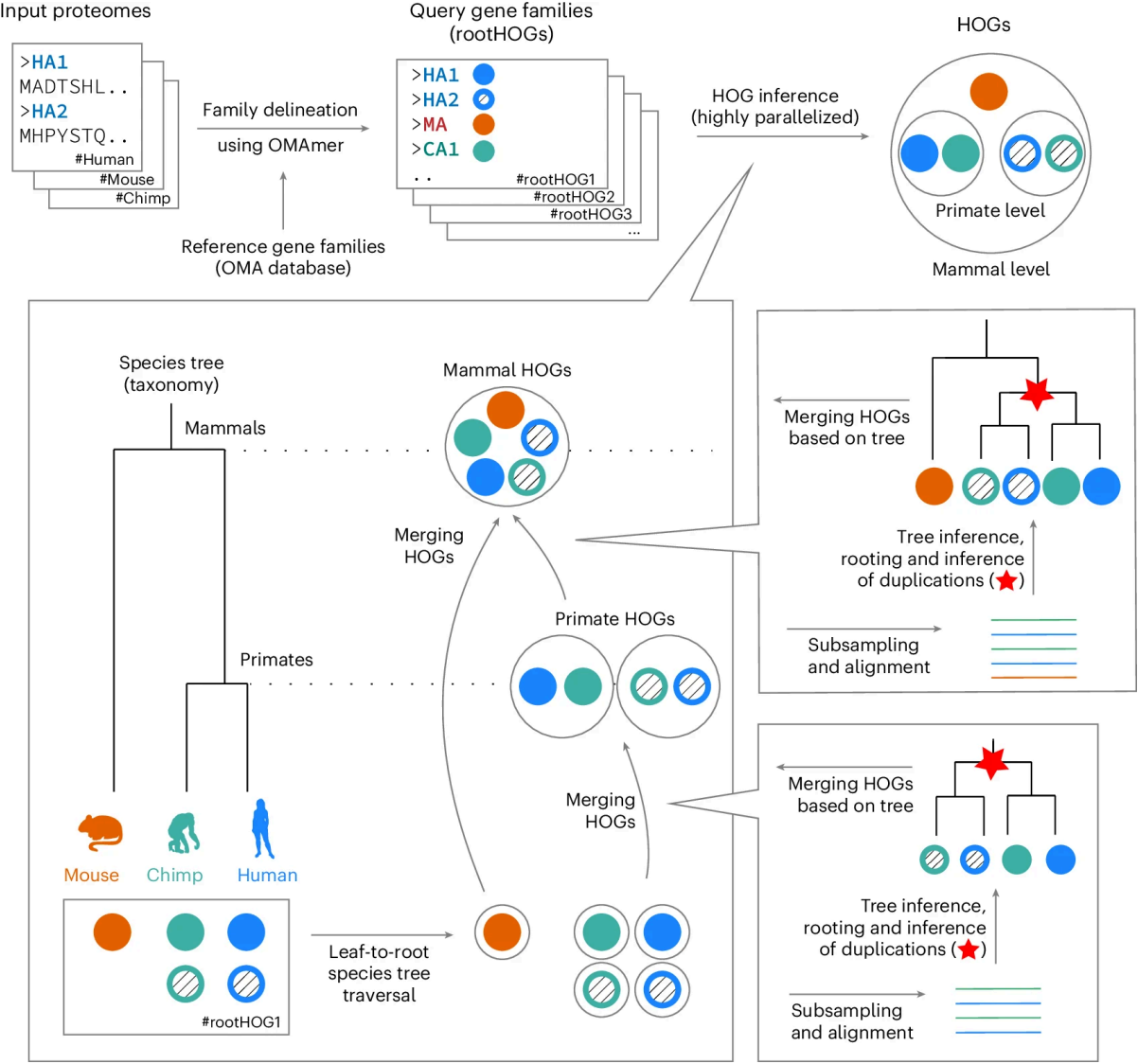
OrthoLogger tools

- `ODB-mapper` — map FASTA file(s) to OrthoDB orthologs
- `orthoLogger` — find orthologs in a set of FASTA files (ab initio)

Protocol

- Query level: `eudicotyledons`
- **OrthoFinder** within-genome pairs × OrthoDB pair table
- Merge on OrthoDB protein IDs → Cartesian join → gene-level pairs

FastOMA



FastOMA

Majidian et al. (2025) *Nature Methods* · omabrowser.org/oma/academy/module/fastOMA

Key concept: Hierarchical Orthologous Groups (HOGs) — nested gene families resolved at each taxonomic level; number of HOGs at a level = gene copies in that ancestral species

Input

- OMAMer DB: `Viridiplantae.h5` or custom
- Rooted species tree (Newick, no branch lengths needed)
- Protein FASTA files (`proteome/` folder)

Step 1 — Family inference

- `OMAMer` (alignment-free k-mer) → places proteins into reference HOGs
- `Linclust` → clusters unplaced proteins into novel families

Step 2 — Orthology inference

- Bottom-up traversal of species tree (leaf → root)
- `MAFFT` → `FastTree2` → speciation/duplication labeling
- Species overlap ratio < 0.1 → speciation → HOGs merged
- Duplication → HOG split

Output - Ortholog groups; `process_ath_-plant` pairs

FastOMA

Custom config:

```
params {  
  write_msas = true  
  report = true  
  write_genetrees = true  
  force_pairwise_ortholog_generation = true  
  filter_method = 'col-elbow-row-threshold'  
  filter_gap_ratio_row = 0.3  
  filter_gap_ratio_col = 0.3  
  nr_repr_per_hog = 12  
  min_sequence_length = 40  
}
```

FastOMA

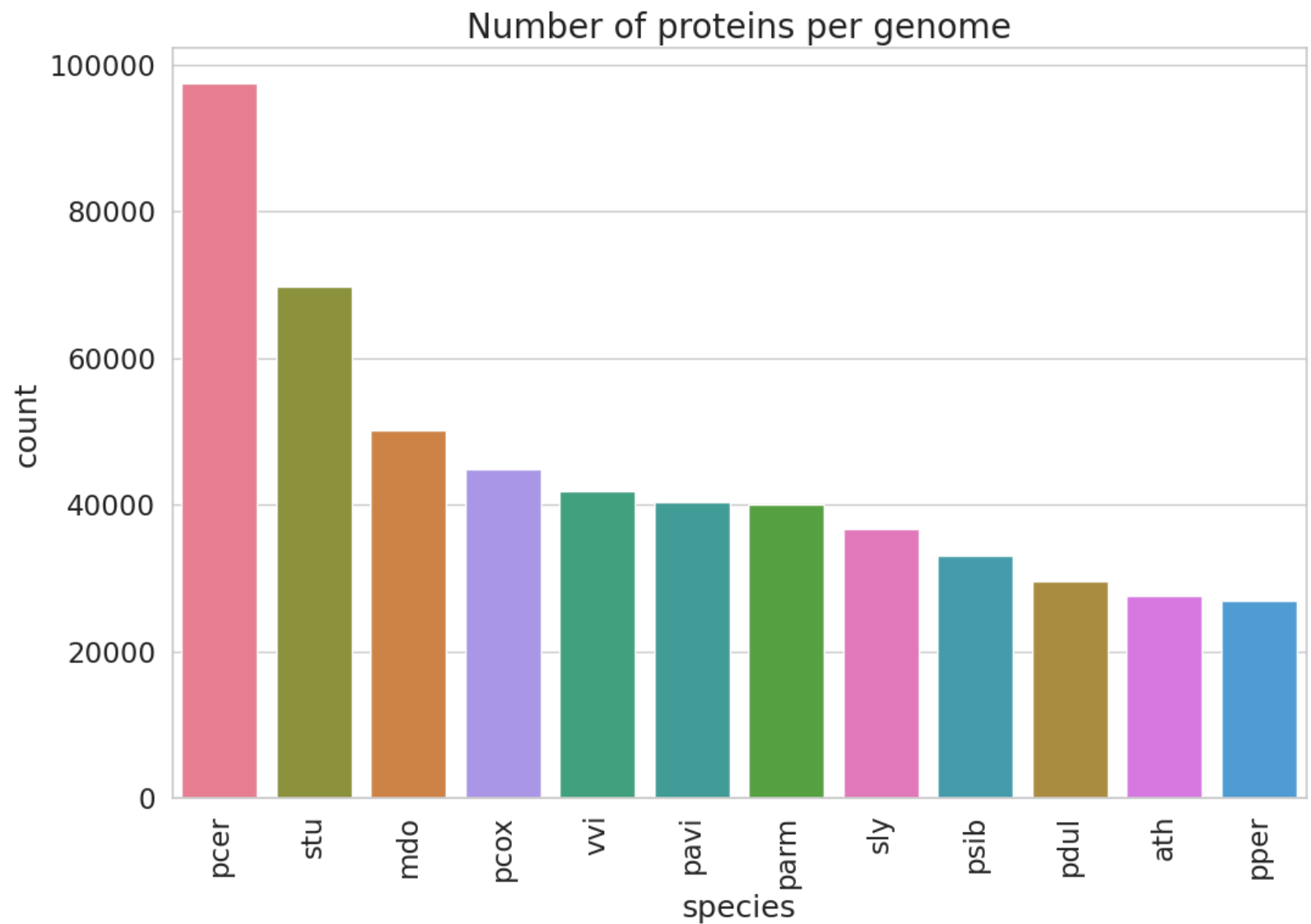
Custom DB:

```
# in ./oma_path/  
wget https://omabrowser.org/All/OmaServer.h5  
wget https://omabrowser.org/All/speciestree.nwk  
  
# NCBITaxonId | ParentTaxonId | Name  
# 1437201 | 91827 | Pentapetalae  
  
omamer mkdb --db Pentapetalae.h5 --oma_path ./oma_path/ \  
--root_taxon "Pentapetalae" --nthreads 28 --log_level info
```

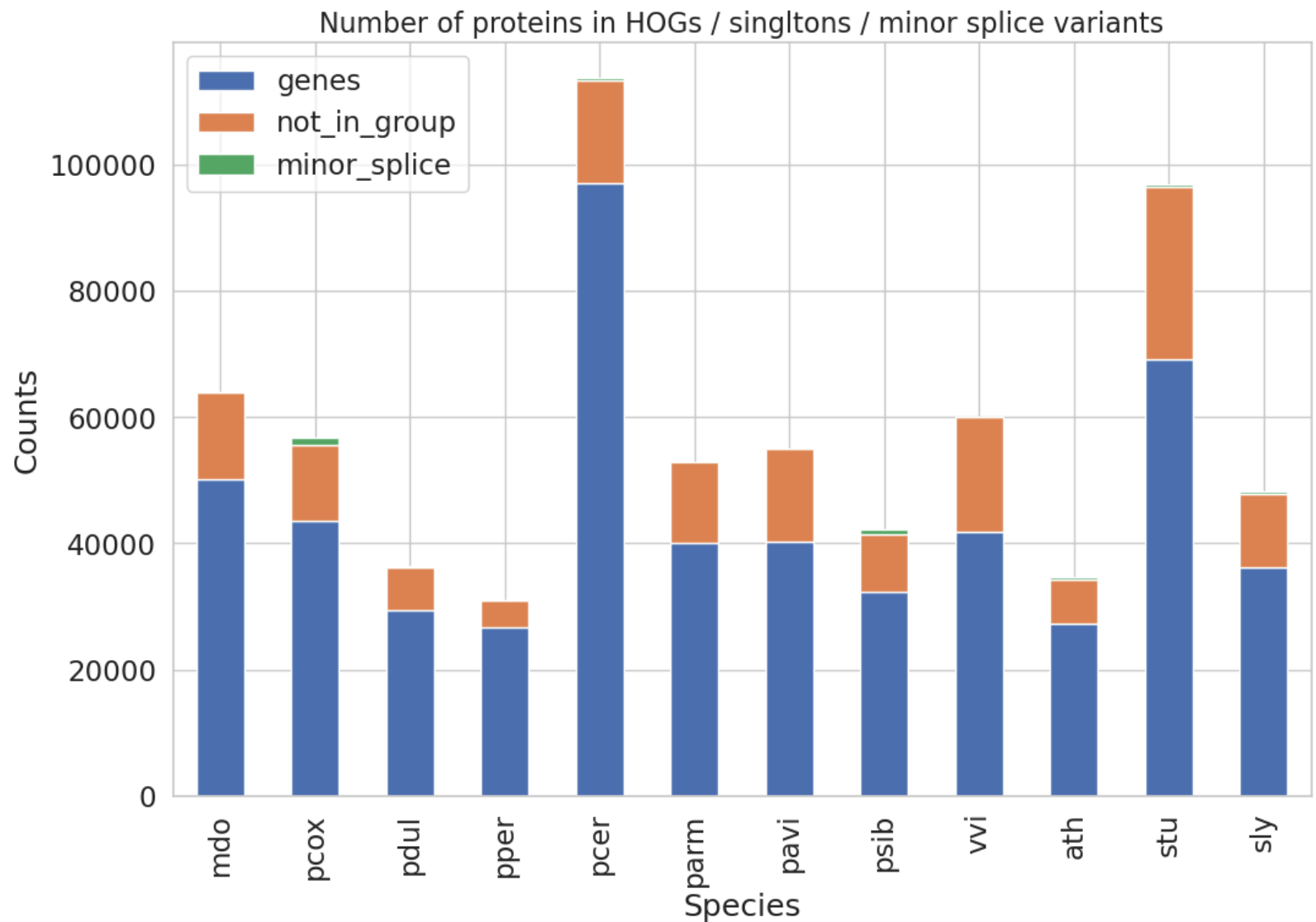
FastOMA

```
nextflow run ../FastOMA/FastOMA.nf -profile standard \  
  --input_folder ./input \  
  --output_folder output \  
  --omamer_db ./input/Pentapetalae.h5 \  
  --max_cpus 24 \  
  --fasta_header_id_transformer noop \  
  --with-report \  
  -c ./input/my_custom.config | tee run_Pentapetalae.log
```

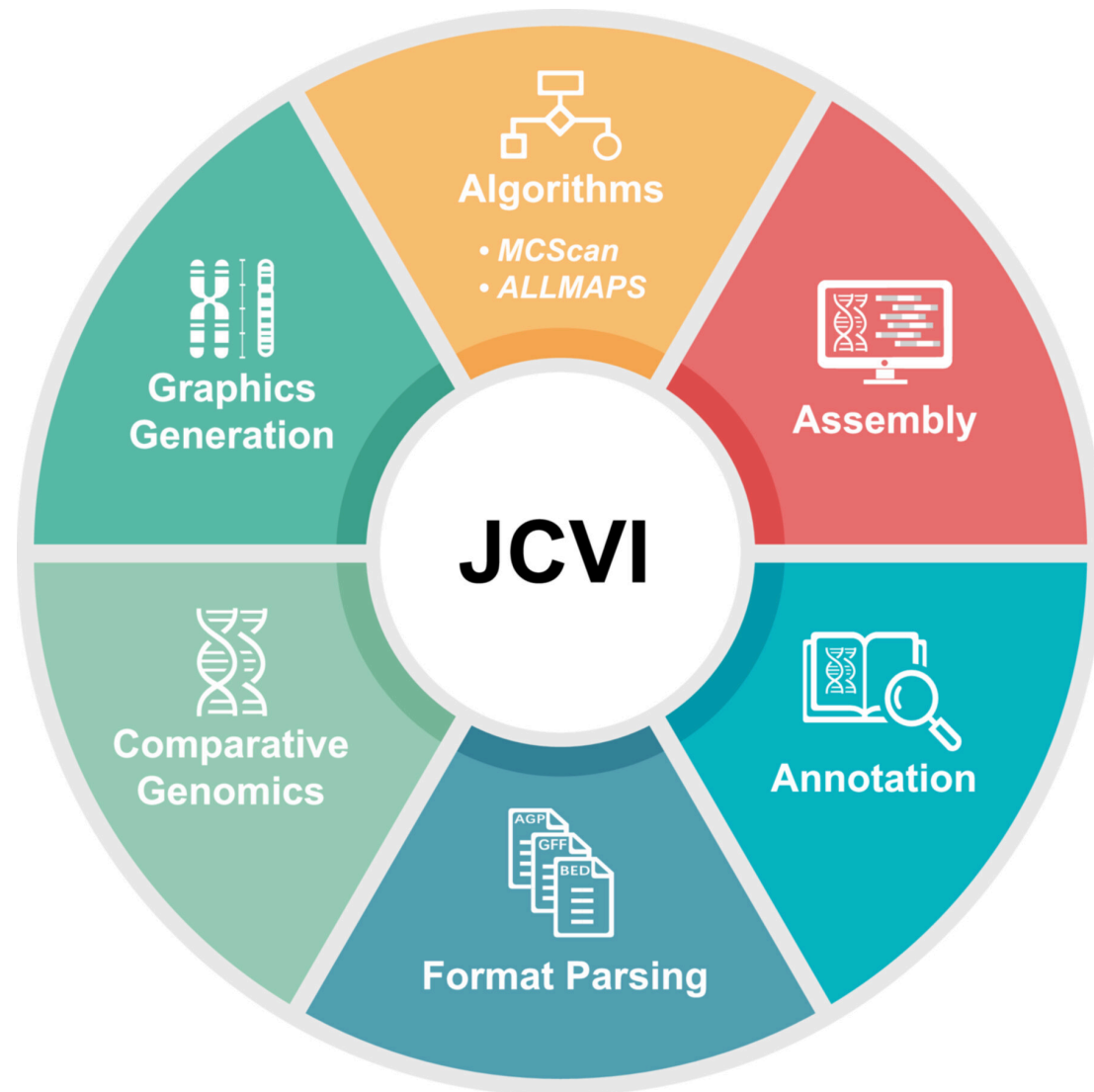

FastOMA



FastOMA



Synteny Analysis



Synteny Analysis — JCVI / MCScanX

Tang et al. (2024) *iMeta*

Pipeline

1. `LAST` — find & align related regions → `.last`
2. Filter: remove tandem duplicates & weak hits → `.last.filtered`
3. Cluster anchors into synteny blocks:
 - `.anchors` — seed blocks (high quality)
 - `.lifted.anchors` — final extended blocks

Notes

- C-score filtering avoids hard E-value cutoffs and repeat artefacts
- Use `--key=Name` or `--key=ID` depending on GFF; `--primary_only` takes first listed isoform — check your GFF

Synteny Analysis — JCVI / MCScanX

```
# create .bed
## keeping one isoform per gene (parameter --primary_only) is problematic,
## since it takes first listed
python -m jcvi.formats.gff bed --type=mRNA --key=Name X.gff3.gz -o X.bed;
# or
python -m jcvi.formats.gff bed --type=mRNA --key=ID X.gff3 -o X.bed;

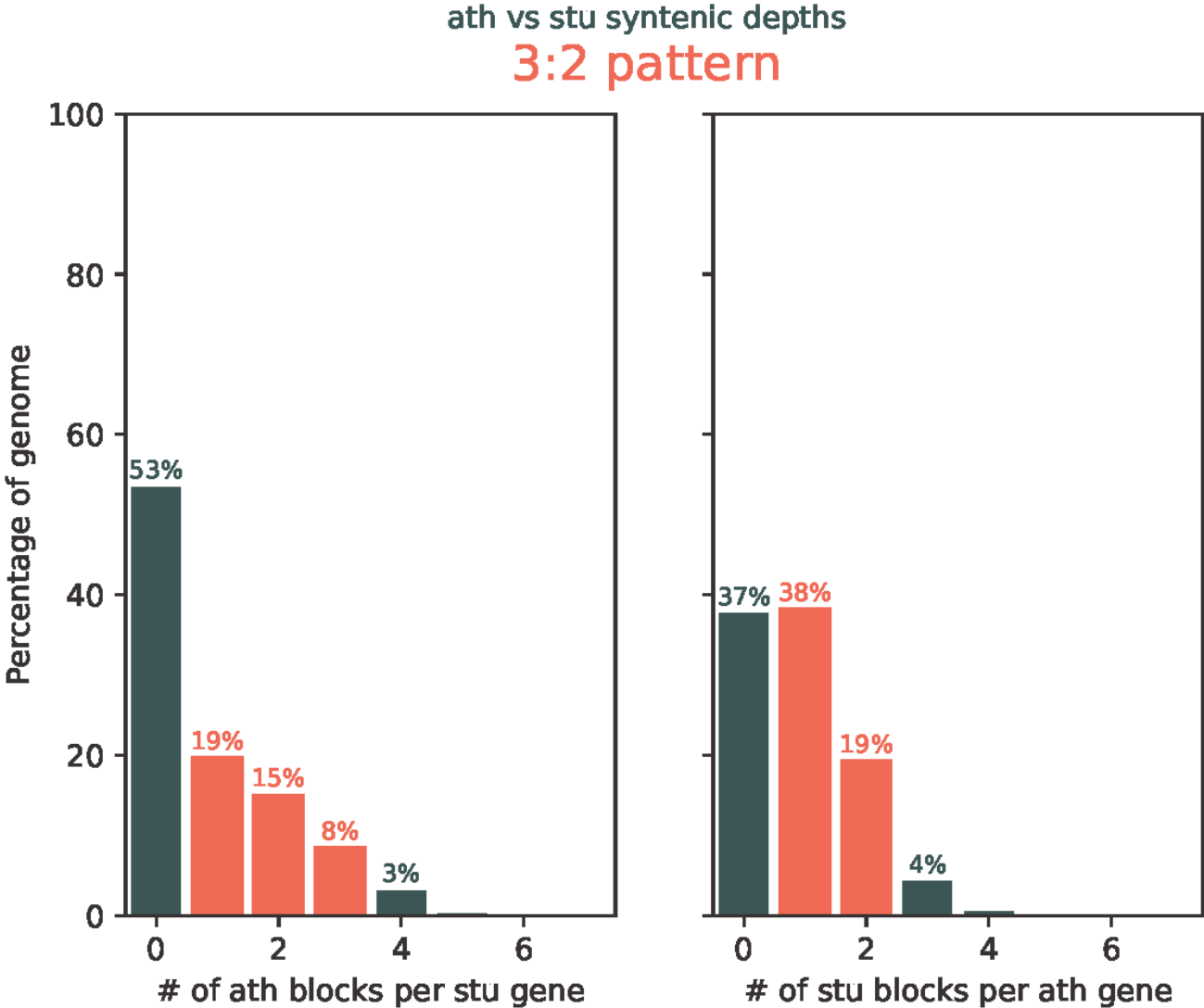
# create .CDS fasta
python -m jcvi.formats.fasta format X.cds.fa.gz X.cds

# run pairwise synteny search
python -m jcvi.compara.catalog ortholog plant1 plant2 --no_strip_names

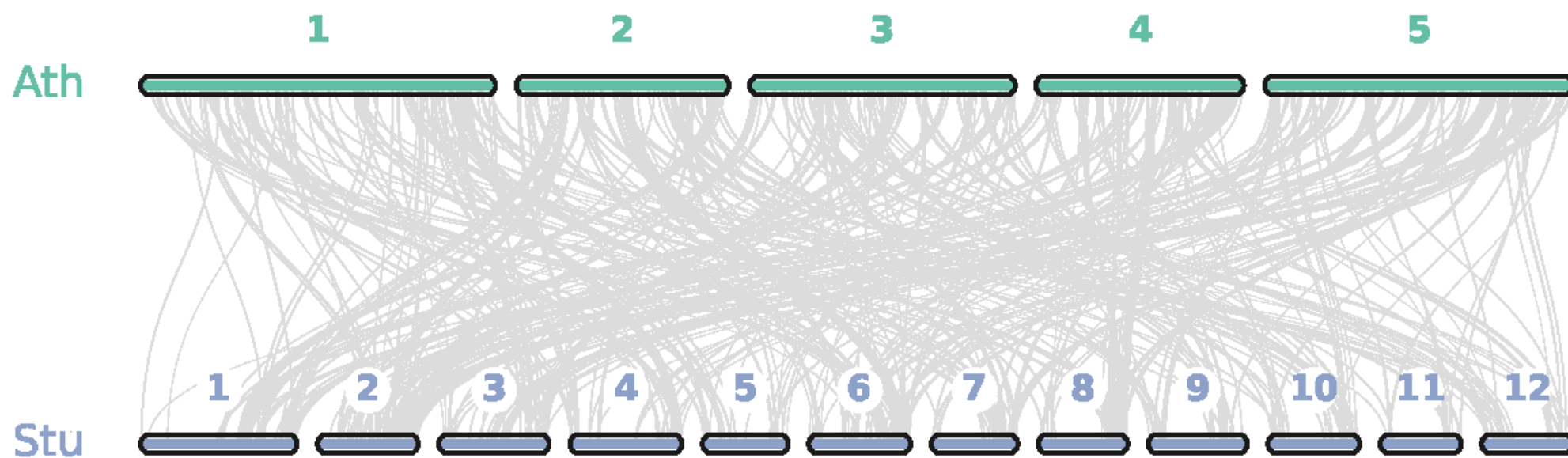
# run pairwise synteny visualization
python -m jcvi.graphics.dotplot plant1.plant2.anchors
python -m jcvi.compara.synteny depth --histogram plant1.plant2.anchors

# macrosynteny visualization
python -m jcvi.compara.synteny screen --minspan=30 \
--simple plant1.plant2.anchors plant1.plant2.anchors.new
python -m jcvi.graphics.karyotype seqids layout
```

Synteny Analysis



Synteny Analysis



Integration & Annotation

Combining multiple methods into a final ortholog set

Integration Strategy — Filter Logic

Results from all six tools combined with a **priority-based filter**:

Priority order:

1. MCScanX ← synteny (highest confidence)
 2. Ensembl Compara
 3. PLAZA
 4. OrthoDB
 5. RBH
 6. FastOMA
-] — require $\geq 2/3$ agreement
] OR MapMan4 corroboration

Rule: once a from_geneID is assigned, it cannot be reassigned
(covered_genes set — prevents double-counting)

Annotation — MapMan4 / Mercator7

Mercator v4.x (web tool) — assigns **MapMan4 BINs**: hierarchical functional classification for plants

Input: per-species protein FASTA → `Y_Mercator4vX_results.txt`

Output columns: IDENTIFIER · BINCODE · NAME · DESCRIPTION

- Normalise IDs + collapse multiple BINs per gene

Note: Each species required custom ID regex normalisation (case fixes, isoform stripping, special translation tables for *stu*)

MapMan4 brings **independent biological context** — a single method's ortholog call becomes credible when both genes share the same functional BIN

Ortholog Validation — Additional Evidence

Independent biological data serve as checkpoints for ortholog calls:

- **qPCR primer sets** — existing primers and expression data confirm orthology
- **Validated genome-scale metabolic models** — functional GEMs confirm genes are assigned to correct enzymatic reactions
- **TF functional cluster databases** — curated databases group transcription factors into functional clusters, providing an independent reference
- **Ontology terms and functional annotations** — shared GO terms, pathway membership, curated functional annotations across species corroborate ortholog assignments

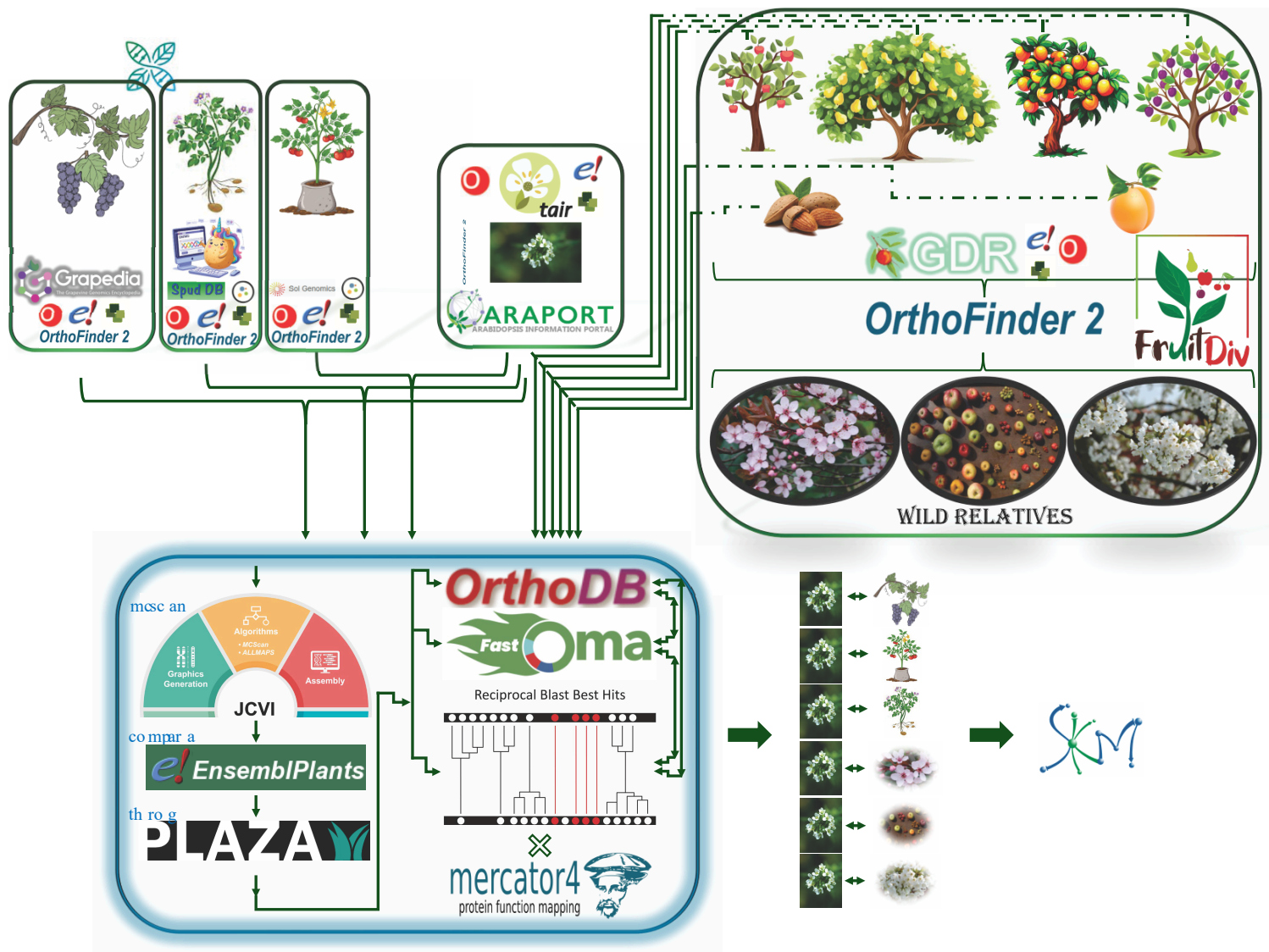
From Ortholog Table to Biological Insight

The final ortholog table feeds directly into downstream biology:

- Orthologs from Arabidopsis carry established pathway membership
- Each gene in potato / fruit trees mapped to knowledge graph nodes
- Enables cross-species network comparison
- Candidate stress/resistance genes identified in wild relatives
- Supports GWAS and QTL candidate gene annotation

Workflow & Summary

Full Pipeline Overview



Summary

Six tools + annotations, one integrated result:

Tool	Type	Priority
JCVI / MCScanX	Synteny	★★★★★
Ensembl Compara	Tree-based	★★★★☆
PLAZA 5.0	Tree-based, plant-specific	★★★★☆
OrthoDB / OrthoLogger	Hierarchical OGs	★★★☆☆
RBH	Sequence similarity	★★★☆☆
FastOMA	HOG-based, scalable	★★★☆☆

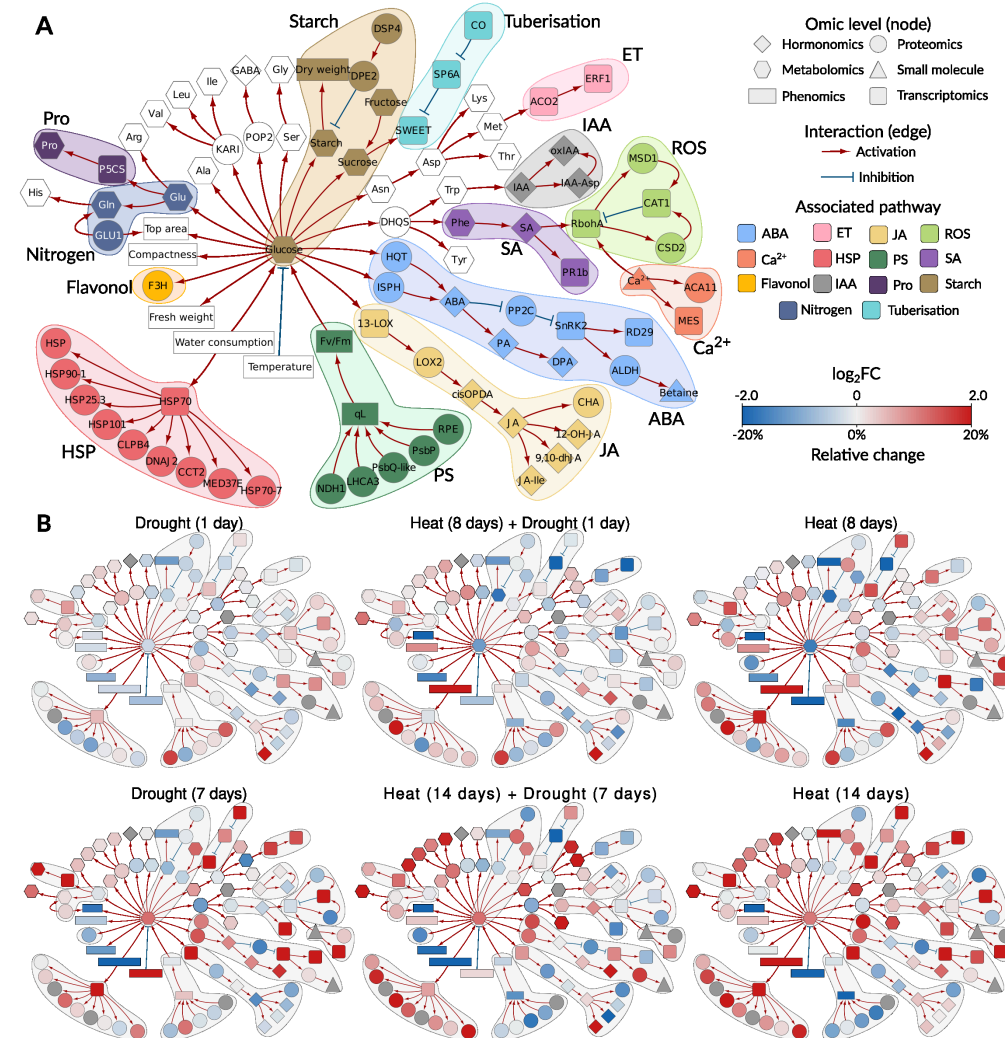
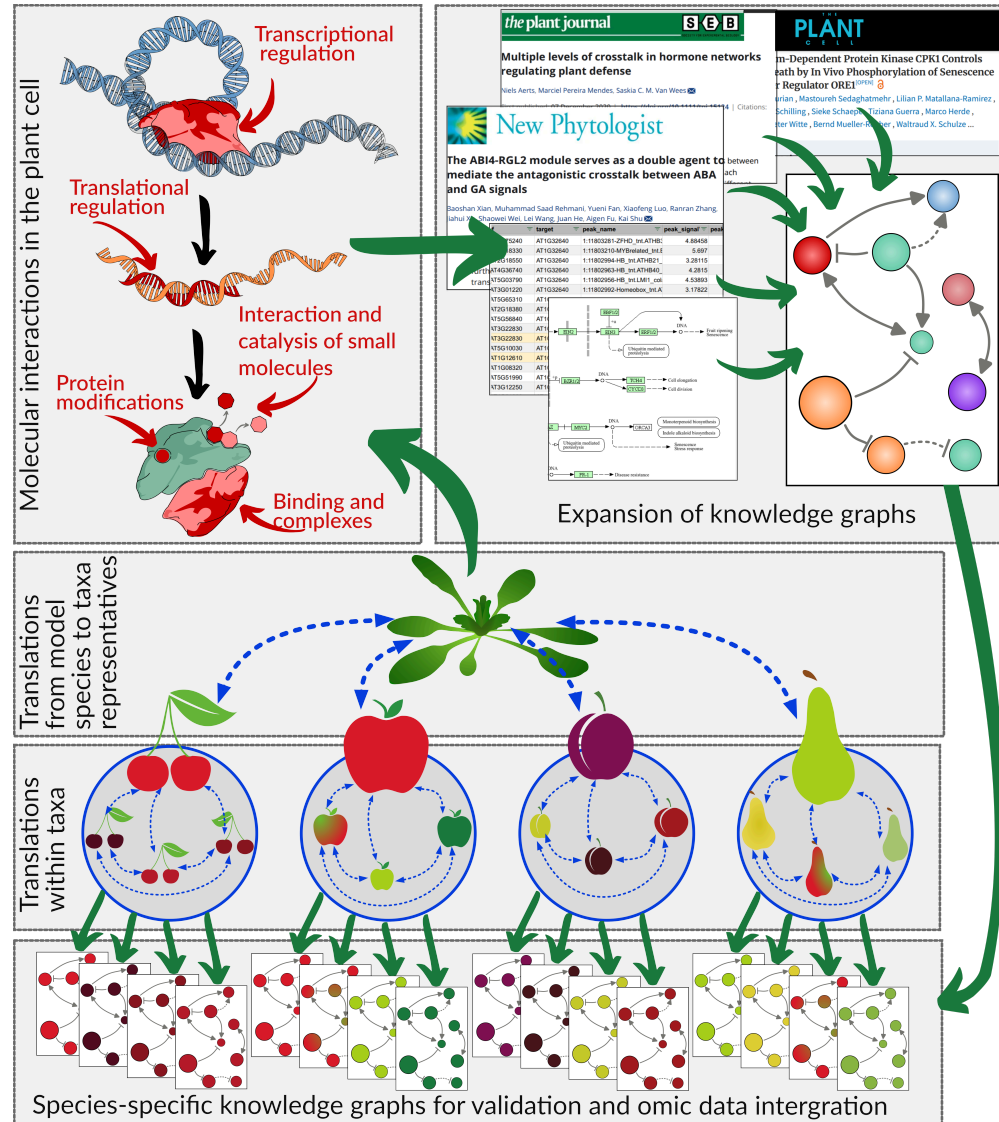
Summary

Why it matters:

- FruitDiv needs reliable orthologs to exploit CWR diversity in *Prunus* / *Malus* / *Pyrus*
- ADAPT needs cross-species SKM connections to transfer potato stress knowledge
- Plant polyploidy makes this hard — synteny + consensus approach handles it robustly

The pipeline is fully scripted in R (`scripts/skm-tools`).
All steps are reproducible.

Summary



Resources & References

Tools

- JCVI/MCScanX: github.com/tanghaibao/jcvi
- OrthoFinder:
github.com/davidemms/OrthoFinder
- FastOMA:
github.com/DessimozLab/FastOMA
- PLAZA: vandepoelelab.be/plaza/
- OrthoDB: orthodb.org
- Mercator7/MapMan4:
plabipd.de/mapman_main.html

Key Papers

- Tang et al. (2024) JCVI. *iMeta*
- Majidian et al. (2025) FastOMA. *Nature Methods*
- Van Bel et al. (2022) PLAZA 5.0. *Nucleic Acids Research*
- Emms & Kelly (2019) OrthoFinder. *Genome Biology*
- Tegenfeldt et al. (2025) OrthoDB v12. *Nucleic Acids Research*